



LGPS: A lightweight GAN-based approach for polyp segmentation in colonoscopy images

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ABSTRACT

Accurate and real-time polyp segmentation is essential for early colorectal cancer detection, yet remains challenging due to small, low-contrast lesions and the computational demands of many deep learning models. We propose LGPS, a Lightweight GAN-based Polyp Segmentation framework that achieves high segmentation accuracy with minimal computational overhead. LGPS integrates a MobileNetV2-based generator enhanced with Residual Squeeze-and-Excitation (ReSE) blocks and a discriminator equipped with Convolutional Conditional Random Fields (ConvCRF) to refine spatial coherence and boundary precision. A hybrid loss function — combining Binary Cross-Entropy, Weighted IoU, and Dice Loss — improves class imbalance handling and sensitivity to small or blurry polyps. LGPS is an extremely compact model, requiring only 1.07 million parameters — over 17× smaller than many recent transformer- and CNN-based SOTA architectures — while preserving high segmentation accuracy. In quantitative evaluation, LGPS achieves a Dice score of 0.7299 and an IoU of 0.7867 on the multi-center PolypGen dataset, the most challenging benchmark in polyp segmentation. On the widely used CVC-ClinicDB dataset, LGPS attains the highest IoU (0.9238) among lightweight and transformer-based approaches. The model also runs at 100.08 FPS on 256 × 256 inputs, demonstrating true real-time capability. Stratified evaluation and qualitative results further confirm its robustness on small and low-contrast lesions. These results highlight LGPS as a computationally efficient yet high-performing framework suitable for real-time clinical deployment. The code is available at <https://github.com/Falmi/LGPS/>.

1. Introduction

Colorectal cancer (CRC) is one of the most prevalent and deadly forms of cancer worldwide, accounting for 10% of all cancer-related deaths [1]. Early detection and removal of polyps during colonoscopy are critical to preventing the progression of CRC. However, manually identifying and segmenting polyps in colonoscopy images is challenging due to their significant variability in size, shape, color, and texture, along with the presence of image artifacts such as motion blur, reflectance, and low contrast with surrounding tissues [2,3]. These challenges often lead to false negatives or inaccurate segmentations, underscoring the need for robust and reliable Computer-Aided Diagnosis (CAD) systems.

In recent years, deep learning-based approaches, particularly Convolutional Neural Networks (CNNs), have shown remarkable success in automating polyp segmentation in colonoscopy images [4,5]. Models such as U-Net [6], PraNet [5], HardNet-MSEG [7], and WDF-Net [8] have achieved remarkable segmentation accuracy by leveraging

encoder-decoder architectures, attention mechanisms, and multi-scale feature fusion. Despite their effectiveness, these methods face three significant limitations: (i) high computational cost, making them unsuitable for real-time applications [9]; (ii) difficulty in segmenting small or low-contrast polyps [10]; and (iii) poor generalization across datasets due to variations in imaging conditions [3]. These limitations hinder the adoption of these segmentation methods in clinical practice.

In addition to these classical CNN-based limitations, recent 2023–2025 models — including transformer-based methods such as SSFormer [11] and TransNetR [12], dual-branch attention architectures such as WDF-Net [8], and emerging Mamba-based segmentation frameworks [13,14] — have improved global feature modeling but still present several critical shortcomings. First, these approaches rely on heavy backbones (e.g., PVT, Swin, or Mamba blocks) with tens of millions of parameters, resulting in substantial computational overhead that restricts real-time use during colonoscopy. Second, despite

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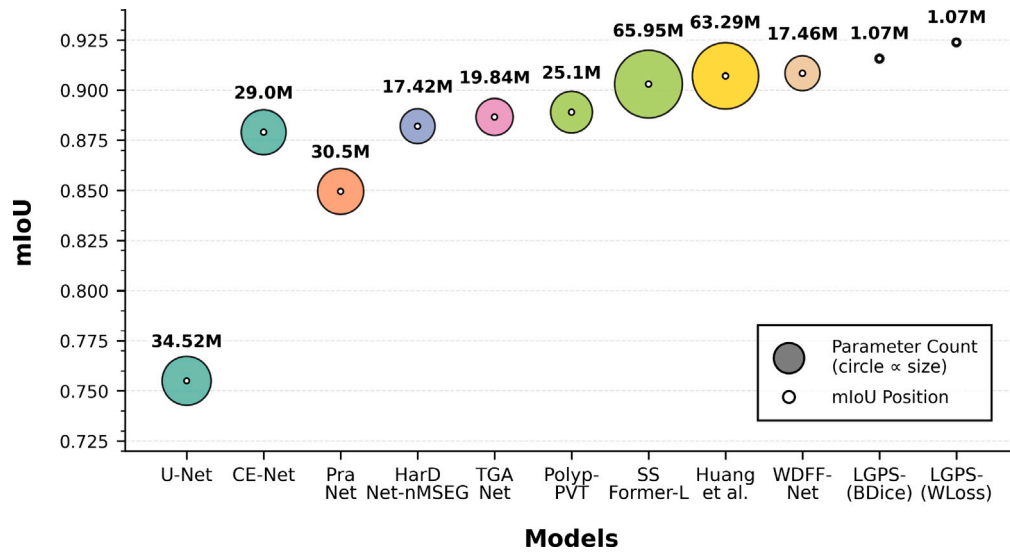


Fig. 1. Comparison of model size and segmentation performance on the CVC-ClinicDB dataset. Each marker represents a polyp segmentation model, where the circle size corresponds to parameter count. LGPS achieves the highest IoU among all methods while using only 1.07M parameters, demonstrating superior efficiency-accuracy trade-off.

their global modeling capability, many of these models continue to underperform on small, low-contrast, or irregular polyps, particularly on challenging multi-center datasets such as PolypGen [3]. Third, their complex encoder-decoder designs exhibit limited robustness under variable illumination, motion blur, and device-dependent imaging conditions common in clinical settings. These persisting challenges underscore the need for a purpose-built framework that balances accuracy, robustness, and real-time efficiency — capabilities that current methods do not jointly satisfy.

To address these challenges, we propose the Lightweight GAN-based Framework for Polyp Segmentation (LGPS), a novel architecture tailored for accurate, real-time polyp detection under clinical constraints. As shown in Fig. 1, LGPS achieves state-of-the-art segmentation accuracy with only 1.07 million parameters, making it 17× smaller than the smallest existing SOTA model. LGPS introduces three distinctive contributions: (1) a GAN-based architecture featuring a ConvCRF-enhanced discriminator, which improves spatial consistency and boundary precision, a novel adaptation not present in previous GAN segmentation models; (2) a hybrid loss function combining Binary Cross-Entropy, Weighted IoU, and Dice Loss, specifically designed to mitigate class imbalance and enhance small-object and low-contrast segmentation; and (3) a MobileNetV2-based generator augmented with residual Squeeze-and-Excitation (ReSE) blocks, providing enhanced feature sensitivity to subtle polyp boundaries while preserving computational efficiency. Unlike transformer- or Mamba-based SOTA methods, LGPS simultaneously offers fast inference (>100 FPS), high segmentation accuracy, strong generalization across datasets, and robustness to small or low-contrast lesions, making it highly suitable for practical clinical deployment.

Beyond architectural design, LGPS is rigorously validated through extensive experiments, including qualitative analysis on small and low-contrast polyps, ablation studies, and cross-dataset evaluations. On the challenging PolypGen dataset, LGPS outperforms larger models in both generalization and efficiency. These results demonstrate that LGPS is not merely a lightweight variant of prior work, but a purpose-built framework that delivers practical, real-time performance for clinically relevant segmentation tasks.

2. Related work

The rapid development of deep learning methods has significantly advanced computer vision, particularly in medical image segmentation. In this section, we provide an overview of major progress in medical

image segmentation methods, focusing on polyp segmentation research. We highlight the strengths and limitations of existing approaches, which emphasize the need for lightweight, efficient, and generalizable models suitable for real-time clinical applications.

2.1. U-Net architectures

The domain of medical image segmentation has made remarkable progress with the advent of deep learning. U-Net [15], introduced in 2015, revolutionized the field with its encoder-decoder architecture and skip connections, enabling precise localization and segmentation of medical structures. Since then, U-Net has become a benchmark for many segmentation tasks due to its simplicity and effectiveness. However, it struggles with complex visual patterns, such as polyps with blurry edges or low contrast, often leading to under-segmentation or over-segmentation [16].

To address these limitations, subsequent works have introduced advanced architectures and techniques. CE-Net [17] enhances U-Net by incorporating a context encoder module to capture global context information, improving segmentation accuracy in challenging regions. Similarly, PraNet [5] incorporates parallel reverse attention modules to focus on boundary cues and region relationships, achieving SOTA performance in polyp segmentation tasks. Despite their improvements, these models often require significant computational resources, making them unsuitable for real-time applications or deployment on resource-constrained environments [7].

2.2. Attention mechanisms and feature aggregation

Attention mechanisms have emerged as a powerful tool for improving segmentation accuracy by enabling models to focus on diagnostically relevant regions [18]. HarDNet-MSEG [7] uses a cascaded partial decoder and the HarDNet68 [19] backbone to achieve high accuracy and inference speed, making it suitable for real-time applications. SANet [20] introduces a shallow attention module to address pixel imbalance in small polyps, effectively reducing background noise and improving segmentation accuracy. TGA-Net [10] leverages text-based embeddings and auxiliary classification tasks to handle drastic scale variations in polyp size. While these methods demonstrate remarkable results, their computational complexity remains a significant drawback [11].

2.3. Transformer-based approaches

The emergence of transformer-based models has further advanced the field of medical image segmentation. Transformers excel at capturing long-range dependencies, making them particularly effective for complex segmentation tasks [21]. Polyp-PVT [9] employs a Pyramid Vision Transformer (PVT) to learn robust feature representations, achieving SOTA performance in polyp segmentation. SSFormer [11] introduces a progressive local decoder to refine segmentation results, while WDFNet [8] combines dual-branch feature fusion with progressive and scale-aware strategies to address under-segmentation and size variation. Despite their effectiveness, transformer-based models are often computationally expensive, with large numbers of parameters and slow inference times, limiting their practicality for real-time applications [22].

2.4. Auto-configuring segmentation models

nnU-Net [23] represents a self-configuring (SConf.) segmentation framework that automatically adapts its preprocessing, network architecture, training, and postprocessing pipelines based on the dataset at hand. Originally proposed for general medical image segmentation, nnU-Net has demonstrated exceptional performance on multiple benchmarks, including polyp segmentation. Its 3D variant processes input patches of size $128 \times 128 \times 128$ and comprises 91.5 million parameters, making it significantly larger and more computationally intensive than most models evaluated in this work. While its plug-and-play nature and high accuracy make it attractive for offline clinical analysis, its substantial memory and computational demands limit its applicability in real-time or embedded scenarios. In contrast, our proposed LGPS is specifically optimized for low-resource deployment, offering a 17× smaller parameter count and real-time inference speed without sacrificing segmentation quality.

2.5. Lightweight models for real-time deployment

Given the computational challenges of existing methods, there is a growing demand for lightweight models that can deliver competitive performance while maintaining low memory and computational requirements, especially in resource-constrained environments [24]. Several works in the literature have proposed lightweight architectures for image segmentation, demonstrating the feasibility of efficient and real-time solutions.

Ni et al. [25] introduced a bilinear attention network with an adaptive receptive field for the segmentation of surgical instruments. Their approach leverages bilinear attention mechanisms to capture fine-grained details while maintaining computational efficiency. Similarly, Wang et al. [26] proposed LEDNet, a lightweight encoder-decoder network that uses ResNet50 in the encoder block and an attention pyramidal network in the decoder block. LEDNet achieves real-time semantic segmentation with a significant reduction in computational complexity, making it suitable for resource-constrained environments. Another notable contribution is Squeeze U-Net [27], which is inspired by the U-Net architecture [15]. Squeeze U-Net achieves a 12× reduction in model size compared to traditional U-Net while maintaining efficient performance in terms of multiplication-accumulation (MAC) operations and memory usage. This makes it particularly suitable for deployment on devices with limited computational resources. ERFNet [28] introduced an efficient residual factorized convolutional network for real-time semantic segmentation. ERFNet achieves a balance between accuracy and speed, making it a strong candidate for real-time applications.

These works highlight the potential of lightweight architectures for real-time image segmentation. However, most existing lightweight models are designed for general-purpose segmentation tasks and have not been specifically optimized for polyp segmentation in colonoscopy images. Polyp segmentation presents unique challenges, such as the

Table 1

Summary of the LGPS generator and discriminator components.

Component	Configuration
Input	256 × 256 × 3 RGB image
Encoder	MobileNetV2 backbone (ImageNet weights)
Skip connections	Multi-scale encoder features (128 × 128, 64 × 64, 32 × 32, 16 × 16) fused with decoder
ReSE blocks	Applied after encoder-decoder feature fusion in each decoder stage
Decoder	4× UpSampling2D (bilinear) + skip concat + ReSE
Output layer	1 × 1 Conv2D + Sigmoid (binary mask)
Discriminator input	Image-mask pair (256 × 256 × 4)
Conv layers	5 layers (filters: 64, 128, 256, 512, 1; stride = 2)
Activation	LeakyReLU ($\alpha = 0.2$) for first 4 layers
ConvCRF	4× conv + sigmoid refinement layers
Inference mode	Discriminator disabled; generator only

need to handle small and irregularly shaped polyps, blurry boundaries, and low contrast with surrounding tissues. These challenges require the development of specialized lightweight models tailored to the particularities of polyp segmentation.

Addressing this significant gap, we propose LGPS, an efficient lightweight GAN-based framework designed specifically for polyp segmentation. LGPS achieves SOTA segmentation accuracy without sacrificing computational efficiency, making it suitable for real-time clinical applications.

3. Proposed model architecture

LGPS is a novel GAN-based architecture designed for efficient and precise polyp segmentation in medical images. The model leverages a lightweight backbone, modified Residual Blocks with Squeeze-and-Excitation (ReSE) [29] mechanisms, and a refinement module to achieve SOTA segmentation accuracy while maintaining computational efficiency. As shown in Fig. 2, the LGPS architecture consists of two primary components: a generator (G) that produces segmentation masks from input images and a discriminator (Dis.) that evaluates the quality of these masks.

Table 1 summarizes the main components and runtime behavior. In the following subsections, we provide more detailed descriptions of the generator and discriminator to enhance clarity and reproducibility.

3.1. Generator architecture

The generator follows an encoder-decoder architecture with a modified MobileNetV2 backbone, chosen for its efficiency and lightweight design. In our implementation, the encoder is derived from a pre-trained MobileNetV2 [30], (ImageNet weights). Four intermediate feature maps are extracted as skip connections to the decoder. In the decoder, ReSE blocks are applied after the fusion of encoder skip features with the upsampled decoder features, ensuring that multi-scale spatial information is refined before further decoding. A final 1 × 1 convolution with sigmoid activation produces the binary segmentation mask. Importantly, only the generator is used during inference; the discriminator and ConvCRF modules act solely as training-time regularizers, so inference relies exclusively on the lightweight 1.07 M-parameter generator.

While Table 1 provides a high-level summary of the LGPS components, Table 2 offers a detailed layer-by-layer specification of the modified MobileNetV2 encoder and ReSE-based decoder to ensure full reproducibility. In LGPS, the original MobileNetV2 expansion factors (6×) and channel widths are reduced by half, all classification-specific layers are removed, and only stages E1–E5 are retained to preserve multi-scale spatial features. These modifications result in a total generator size of 1.07M parameters, verified using the TensorFlow model summary. The components of the generator are as follows:

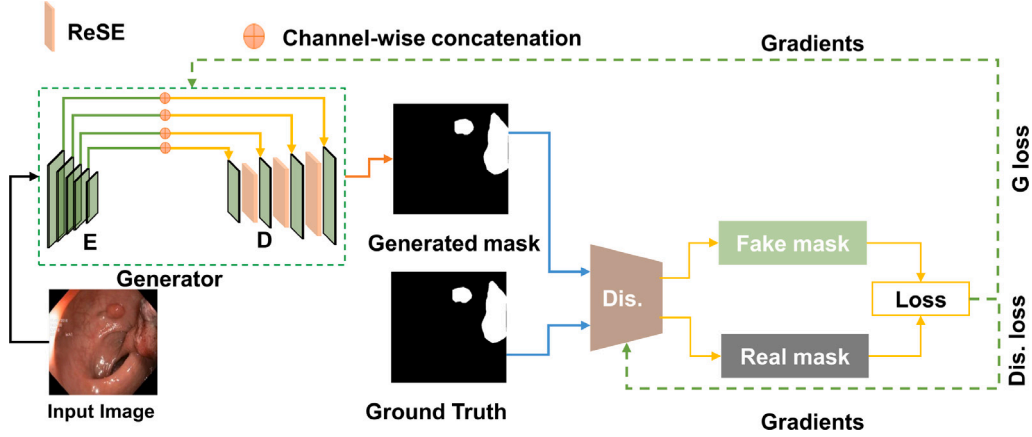


Fig. 2. Overview of the proposed LGPS architecture. The generator uses a MobileNetV2 encoder whose multi-scale features are fused with decoder features, followed by ReSE refinement after each fusion step. The decoder progressively upsamples these refined features to produce the segmentation mask. The discriminator is a convolutional network with ConvCRF refinement used only during training; inference uses the generator alone.

Table 2

Layer-by-layer configuration of the modified MobileNetV2-ReSE generator used in LGPS. ReSE blocks are applied only after encoder-decoder feature fusion in the decoder.

Stage	Operation	Output size	Notes
Input	–	$256 \times 256 \times 3$	–
E1	Conv 3×3 , 16 filters Inverted Bottleneck $\times 1$	$128 \times 128 \times 16$ $128 \times 128 \times 16$	Stride = 2 Expansion = 2; skip 1
E2	Inverted Bottleneck $\times 2$	$64 \times 64 \times 24$	Halved channels; skip 2
E3	Inverted Bottleneck $\times 3$	$32 \times 32 \times 32$	Reduced channels; skip 3
E4	Inverted Bottleneck $\times 3$	$16 \times 16 \times 64$	Deeper features; skip 4
E5	Inverted Bottleneck $\times 1$	$8 \times 8 \times 96$	Bottleneck encoder output
D1	Upsample $\times 2$ + concat(E4) ReSE Block	$16 \times 16 \times (96 + 64)$ $16 \times 16 \times 64$	Decoder stage 1 Applied after fusion
D2	Upsample $\times 2$ + concat(E3) ReSE Block	$32 \times 32 \times (64 + 32)$ $32 \times 32 \times 32$	Decoder stage 2 Applied after fusion
D3	Upsample $\times 2$ + concat(E2) ReSE Block	$64 \times 64 \times (32 + 24)$ $64 \times 64 \times 24$	Decoder stage 3 Applied after fusion
D4	Upsample $\times 2$ + concat(E1) ReSE Block	$128 \times 128 \times (16 + 24)$ $128 \times 128 \times 16$	Decoder stage 4 Applied after fusion
Output	Conv 1×1 + Sigmoid	$256 \times 256 \times 1$	Segmentation output

3.1.1. Encoder

The encoder is built using a pre-trained MobileNetV2 model, which employs depthwise separable convolutions to reduce computational complexity while maintaining performance. To further optimize the model, we made the following modifications to MobileNetV2: (i) reduce the number of filters in each layer by a factor of 2 to decrease the number of parameters while retaining essential feature extraction capabilities; (ii) remove the final classification layer and redundant intermediate layers to keep only the essential feature extraction layers for segmentation tasks; and (iii) add depthwise separable convolutions with reduced expansion factors to minimize computational overhead.

These modifications resulted in a lightweight MobileNetV2 backbone with only 1.07 million parameters. The encoder extracts feature maps from four intermediate layers, which are used as skip connections to preserve spatial details during decoding. This multi-scale feature extraction enables the model to handle polyps of varying sizes and shapes, including small polyps that are often missed by other methods.

3.1.2. Modified Residual with Squeeze-and-Excitation block (ReSE)

The ReSE block is an essential component of the generator architecture, aiming to enhance feature extraction and recalibration. As shown in Fig. 3(a), it combines traditional residual connections with the SE mechanism, enabling the model to dynamically recalibrate feature maps and focus on diagnostically relevant regions.

The ReSE block consists of several components that work together to improve feature representation and segmentation performance. They are described as follows.

The first component is the bottleneck layer, which reduces the number of channels by a factor of 4 using a 1×1 convolution layer. This step reduces computational complexity while preserving essential features. The output of the 1×1 convolution is passed through a Batch Normalization layer and a ReLU activation function, expressed as:

$$x_{\text{bottleneck}} = \text{ReLU}(\text{BatchNorm}(\text{Conv2D}_{1 \times 1}(x))), \quad (1)$$

where x is the input tensor, and $\text{Conv2D}_{1 \times 1}$ denotes a 1×1 convolution.

Next, the bottleneck output is passed through a 3×3 convolution layer to extract spatial features. This layer captures local patterns and structures in the feature maps, which are critical for accurate polyp segmentation. The output of the 3×3 convolution is normalized using Batch Normalization:

$$x_{\text{spatial}} = \text{BatchNorm}(\text{Conv2D}_{3 \times 3}(x_{\text{bottleneck}})). \quad (2)$$

A residual connection is then added between the input and the output of the 3×3 convolution to facilitate gradient flow and improve training stability [31]. If the number of channels in the input tensor does not match the output tensor, a 1×1 convolution is applied to the input tensor to adjust its dimensions:

$$x_{\text{shortcut}} = \text{BatchNorm}(\text{Conv2D}_{1 \times 1}(x)). \quad (3)$$

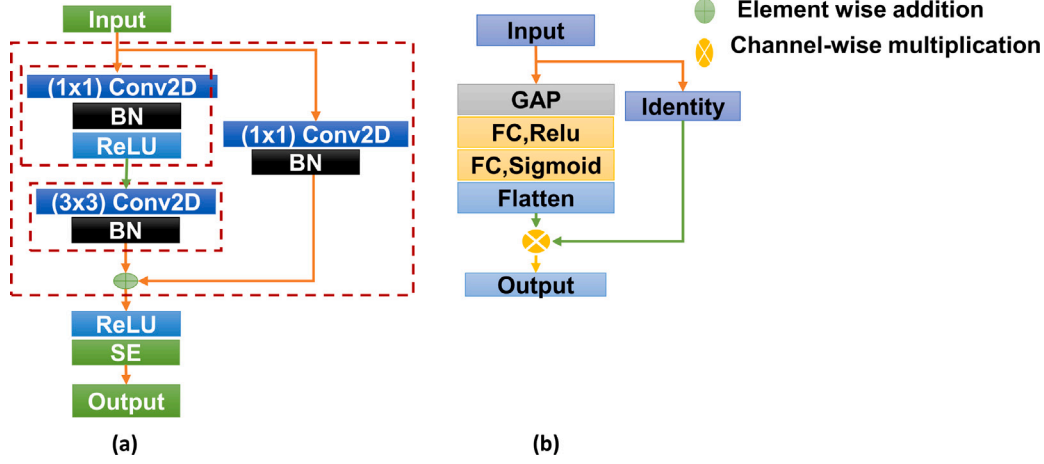


Fig. 3. Structure of the ReSE block. (a) Modified residual block with bottleneck, 3×3 convolution, and identity mapping. (b) Integrated SE module that recalibrates channel responses to enhance low-contrast and boundary-level features.

The residual connection is implemented using an Add operation, followed by a ReLU activation:

$$x_{\text{residual}} = \text{ReLU}(\text{Add}([x_{\text{shortcut}}, x_{\text{spatial}}])). \quad (4)$$

Finally, the Squeeze-and-Excitation (SE) block recalibrates the feature maps by modeling channel-wise interdependencies [29]. As shown in Fig. 3(b), the SE mechanism consists of three steps. First, the squeeze block uses global average pooling to aggregate spatial information into channel-wise descriptors:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j), \quad (5)$$

where $x_c(i, j)$ is the feature map at spatial location (i, j) for channel c , and $H \times W$ is the spatial dimension. Second, the excitation block uses two fully connected (FC) layers to model non-linear channel interdependencies, generating attention weights:

$$s_c = \sigma(W_2 \delta(W_1 z_c)), \quad (6)$$

where W_1 and W_2 are learnable weights, δ is the ReLU activation, and σ is the sigmoid activation. Third, the recalibration block applies the attention weights s_c to the input feature maps using a Multiply operation, emphasizing diagnostically relevant features:

$$x_{\text{se}} = \text{Multiply}([x_{\text{residual}}, s]). \quad (7)$$

This ReSE block, with its combination of bottleneck layers, spatial feature extraction, residual connections, and SE mechanisms, significantly enhances the generator's ability to extract and recalibrate features for accurate polyp segmentation.

3.1.3. Decoder

The decoder progressively upsamples the feature maps and concatenates them with skip connections from the encoder to recover spatial details. It consists of four upsampling stages, each followed by the ReSE. The upsampling is performed using bilinear interpolation, and the skip connections are concatenated with the upsampled feature maps to preserve spatial information. The final output of the decoder is a binary segmentation map, obtained by applying a 1×1 convolution with a sigmoid activation:

$$M_{\text{pred}} = \sigma(\text{Conv2D}(1, (1, 1))(F_d^4)), \quad (8)$$

where F_d^4 is the final feature map from the decoder. The use of skip connections and progressive upsampling ensures that the model preserves fine-grained spatial details, a key strength for accurate polyp segmentation.

3.2. Discriminator architecture

The discriminator employs a patch-based adversarial framework with Convolutional Conditional Random Fields (ConvCRF) refinement to improve spatial consistency in polyp segmentation. It processes concatenated pairs of input images and predicted masks $(I_{\text{in}}, M_{\text{pred}}) \in \mathbb{R}^{256 \times 256 \times 4}$ through five convolutional layers. Each layer uses a kernel size of $(3, 3)$, stride 2, and LeakyReLU activation ($\alpha = 0.2$), progressively increasing the number of filters from 64 to 512. The final layer produces a patch-wise real/fake probability map, providing fine-grained feedback to the generator.

To address spatial inconsistency in GAN-based segmentation, we introduce ConvCRF layers, which refine local spatial coherence through learnable 3×3 convolutions. A ConvCRF layer consists of a 3×3 convolutional operation followed by a sigmoid activation:

$$\text{ConvCRF}(F_d^i) = \sigma(\text{Conv}_{3 \times 3}(F_d^i)), \quad (9)$$

where F_d^i represents the feature maps from the previous layer, and σ is the sigmoid activation function. This operation enforces smoothness in predicted masks while preserving edges.

In our implementation, four ConvCRF layers are applied sequentially after the final convolutional layer for refinement. The refined feature maps are computed as:

$$F_{\text{refined}} = \text{ConvCRF}(F_d^i), \quad (10)$$

where F_d^i represents the feature maps from the final convolutional layer. This ensures smoothness and edge preservation in predicted masks.

The final output is a patch-wise real/fake probability map:

$$D(x) = \sigma(F_{\text{refined}}), \quad (11)$$

where $D(x)$ represents the discriminator's output probability. The discriminator is trained to distinguish between real (ground truth) and generated masks, providing adversarial feedback to the generator. The discriminator (including the ConvCRF refinement layers) is only active during training as an adversarial and spatial regularizer; during inference it is disabled, and the final segmentation mask is produced directly by the generator.

3.3. Adversarial training and loss functions

The generator and discriminator are trained in an adversarial manner, where the generator aims to minimize the difference between real and generated masks, while the discriminator attempts to correctly

classify real and fake masks. The training process follows a minimax game, defined as:

$$\min_G \max_D \mathcal{L}_{\text{total}}(G, D), \quad (12)$$

where G and D represent the generator and discriminator, respectively. The generator's loss function is a weighted combination of Binary Crossentropy (BCE) Loss, Weighted Intersection over Union (WIoU) Loss, and Dice Loss:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{WIoU}} + \lambda_3 \mathcal{L}_{\text{Dice}}. \quad (13)$$

These losses guide the generator to produce accurate and realistic segmentation masks. Below, we describe each component of the hybrid loss function in detail.

3.3.1. BCE Loss

The BCE Loss measures the pixel-wise difference between the predicted mask M_{pred} and the ground truth mask M_{true} . It is defined as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[M_{\text{true}}^i \log(M_{\text{pred}}^i) + (1 - M_{\text{true}}^i) \log(1 - M_{\text{pred}}^i) \right], \quad (14)$$

where N is the total number of pixels, and M_{true}^i and M_{pred}^i are the ground truth and predicted values for the i th pixel, respectively.

3.3.2. WIoU Loss

The WIoU Loss addresses class imbalance by assigning different weights to the foreground (polyps) and background regions. It is defined as:

$$\mathcal{L}_{\text{IoU}} = 1 - \text{WIoU}(M_{\text{true}}, M_{\text{pred}}), \quad (15)$$

where the WIoU is computed as:

$$\text{WIoU}(M_{\text{true}}, M_{\text{pred}}) = \alpha \cdot \text{IoU}_{\text{fg}} + (1 - \alpha) \cdot \text{IoU}_{\text{bg}}. \quad (16)$$

Here, α is the weight for the foreground (typically set to 0.7), and IoU_{fg} and IoU_{bg} are the IoU values for the foreground and background, respectively. These are computed as:

$$\text{IoU}_{\text{fg}} = \frac{\sum(M_{\text{true}} \cdot M_{\text{pred}}) + \epsilon}{\sum M_{\text{true}} + \sum M_{\text{pred}} - \sum(M_{\text{true}} \cdot M_{\text{pred}}) + \epsilon}, \quad (17)$$

$$\text{IoU}_{\text{bg}} = \frac{A + \epsilon}{B + C - A + \epsilon}, \quad (18)$$

where $A = \sum((1 - M_{\text{true}}) \cdot (1 - M_{\text{pred}}))$, $B = \sum(1 - M_{\text{true}})$, $C = \sum(1 - M_{\text{pred}})$, and ϵ is a small constant (e.g., 10^{-6}) to avoid division by zero.

The WIoU loss ensures that the model focuses on both foreground and background regions, addressing the challenge of class imbalance.

3.3.3. Dice Loss

The Dice Loss measures the overlap between the predicted mask and the ground truth mask. It is defined as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum(M_{\text{true}} \cdot M_{\text{pred}}) + \epsilon}{\sum M_{\text{true}} + \sum M_{\text{pred}} + \epsilon}, \quad (19)$$

where ϵ is a small constant to ensure numerical stability. The Dice Loss is particularly effective for segmentation tasks with imbalanced classes, as it emphasizes the overlap between the predicted and ground truth masks.

Table 3

Performance evaluation metrics.

Metrics	Description
Dice	Dice = $(2 \times \text{TP}) / (2 \times \text{TP} + \text{FP} + \text{FN})$
IoU	IoU = $(\text{TP}) / (\text{TP} + \text{FP} + \text{FN})$
Recall	Recall = $(\text{TP}) / (\text{TP} + \text{FN})$
Pre.	Pre. = $(\text{TP}) / (\text{TP} + \text{FP})$
Accuracy	Accuracy = $(\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
F2	F2 = $(5 \times \text{P} \times \text{R}) / (4 \times \text{P} + \text{R})$

3.3.4. Total loss

The total loss for the generator is a weighted combination of the BCE, WIoU, and Dice Losses:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{WIoU}} + \lambda_3 \mathcal{L}_{\text{Dice}}. \quad (20)$$

The weights λ_1 , λ_2 , and λ_3 are hyperparameters that balance the contributions of each loss term. In our experiments, we empirically set $\lambda_1 = 0.4$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.3$ based on preliminary experiments to balance pixel-wise accuracy, segmentation overlap quality, and boundary precision. These loss weights were selected based on exploratory validation experiments conducted on the same validation split used for model selection, and the same configuration was applied across all datasets without dataset-specific tuning. We also observed that LGPS maintains stable performance across a range of reasonable λ combinations, indicating that the architectural components — such as ReSE fusion, adversarial regularization, and ConvCRF refinement — play a much larger role in performance than fine-grained loss-weight adjustments.

3.3.5. Discriminator loss

The discriminator is trained using BCE to classify real and fake masks:

$$\mathcal{L}_{\text{Dis}}(y_{\text{true}}, y_{\text{pred}}) = -\frac{1}{N} \sum_{i=1}^N \left(y_{\text{true}}^i \log(\text{Dis}(y_{\text{pred}}^i)) + (1 - y_{\text{true}}^i) \log(1 - \text{Dis}(y_{\text{pred}}^i)) \right), \quad (21)$$

where y_{true} and y_{pred} are the ground truth and predicted labels, respectively, and Dis is the discriminator's output probability. The adversarial training framework encourages the generator to produce precise and realistic segmentation masks.

4. Experimental result and analysis

4.1. Dataset and evaluation metrics

The experiments utilize five public polyp segmentation datasets: Kvasir-SEG [32], CVC-ClinicDB [33], ETIS [34], CVC-300 [16], and PolypGen [3]. These datasets vary in terms of the number of images and their resolutions. Kvasir-SEG contains 1000 images with variable sizes, while CVC-ClinicDB provides 612 images at a fixed resolution of 384×288 . ETIS consists of 196 images at a higher resolution of 1225×966 . CVC-300 offers 60 images with the same resolution 574×500 , and PolypGen, the largest dataset, contains 1537 images with variable sizes. These datasets collectively provide a diverse and comprehensive foundation for the experiments. All datasets consist of individual colonoscopy frames extracted from video sequences rather than continuous time series, and the acquisition frame rate or sampling frequency is not explicitly provided by the original authors. Consequently, our experiments operate on static images, which is consistent with standard practice in polyp segmentation literature.

To ensure a fair and reproducible evaluation, we follow the commonly adopted protocol in PraNet [5] and related works: a combined training set is formed from Kvasir-SEG and CVC-ClinicDB, while the remaining images from these datasets, along with ETIS, CVC-300, and

PolypGen, are used solely for testing. This multi-dataset evaluation allows us to assess both in-distribution performance (on Kvasir-SEG and CVC-ClinicDB) and out-of-distribution generalization (on ETIS, CVC-300, and PolypGen).

The performance of the LGPS model was evaluated using a Dice coefficient (Dice), Intersection over Union (IoU), Recall, Precision (Pre.) and Accuracy. In addition, we report the F2 score to emphasize recall (sensitivity) in our evaluation. The F2 score weights recall twice as much as precision, which is especially relevant in medical image segmentation tasks like polyp detection, where false negatives (missed polyps) are clinically more critical than false positives. This makes F2 a valuable metric for evaluating models intended for diagnostic support. The formulas for calculating each metric are shown in Table 3. Following prior work in polyp segmentation, we report deterministic performance on fixed test splits rather than p-values or confidence intervals, and we complement quantitative metrics with stratified analysis by polyp size and extensive qualitative comparisons (Sections 4.3.2–4.4).

4.2. Implementation details

The proposed LGPS model is implemented using the TensorFlow and Keras frameworks. The model is trained on an NVIDIA RTX A6000 GPU. The Adam optimizer is used with a learning rate of 1×10^{-4} and a batch size of 16. Input images are preprocessed by resizing them to a fixed resolution of 256×256 pixels and normalizing pixel values to the range $[0, 1]$. To improve the robustness of the model, several data augmentation techniques are applied during training. These include random horizontal and vertical flips with a probability of 0.5, random rotation by an angle between -10° and 10° , random brightness adjustment by a factor between 0.9 and 1.1, and random contrast adjustment by a factor between 0.9 and 1.1. The testing set is not augmented and is directly resized to 256×256 . We chose the 256×256 resolution to balance accuracy and inference speed, particularly for deployment on low-resource devices. All datasets are preprocessed using the same pipeline to ensure that reported performance differences arise from model design rather than dataset-specific preprocessing. Following PraNet [5], 900 and 550 images from the Kvasir-SEG and CVC-ClinicDB datasets are used as the training set, while the remaining 100 and 62 images are used as the testing set.

To ensure full reproducibility, all training configurations are reported here. LGPS is trained for 200 epochs using the Adam optimizer with $\beta_1 = 0.5$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$, and gradient clipping through $\text{clipvalue} = 1.0$. Weight decay is not used. A step-based learning-rate scheduler reduces the learning rate by 10% after epoch 50, and an additional adaptive reduction factor of 0.5 is applied whenever the validation loss fails to improve for three consecutive epochs (minimum learning rate $= 10^{-6}$). Training follows a 1:1 update schedule, with one discriminator and one generator update per iteration. One-sided label smoothing is applied to the discriminator (real labels = 0.9). Gradient accumulation is not used in any experiment.

Model selection is based strictly on validation loss. Although the training schedule spans 200 epochs, the best-performing generator typically emerges much earlier (e.g., around epoch 37 in our experiments), and the model checkpoint with the lowest validation loss is restored at the end of training. During inference, only the generator is active; the discriminator and ConvCRF modules are disabled, so the deployment-time computational cost corresponds solely to the lightweight 1.07 M-parameter generator. The full training script, dataset preprocessing code, and checkpoint files are available in the public repository linked in the abstract.

Inference efficiency was evaluated using only the generator in evaluation mode. All FPS measurements were conducted on an NVIDIA RTX A6000 GPU with input resolution fixed at 256×256 and batch size set to 1. The reported inference speed (100.08 FPS) corresponds to a direct forward-pass measurement of the trained generator. For competing

Table 4

Quantitative evaluation of different loss function configurations on the Kvasir-SEG dataset.

Loss Fun.	Dice	IoU	Recall	Pre.	F2	Acc.
WIoU	0.8530	0.8436	0.7900	0.9123	0.8118	0.9498
BCE_only	0.8552	0.8464	0.7866	0.9202	0.8101	0.9506
Dice_only	0.8494	0.8407	0.7722	0.9494	0.7990	0.9494
BWIoU	0.8515	0.8396	0.8145	0.8816	0.8271	0.9477
BloU	0.8529	0.8431	0.7954	0.9052	0.8152	0.9494
BDice	0.8582	0.8478	0.8049	0.9063	0.8233	0.9508
3Loss A	0.8575	0.8477	0.7905	0.9217	0.8136	0.9512
3Loss B	0.8431	0.8331	0.7880	0.8816	0.8141	0.9457
3Loss C	0.8377	0.8289	0.7775	0.8952	0.7986	0.9450

methods, FPS values are adopted directly from the original publications, as unified re-evaluation on identical hardware is infeasible due to the lack of publicly available implementations and pretrained models. To provide a hardware-independent comparison, Table 7 additionally reports FLOPs for LGPS (1.34 GFLOPs, measured using TensorFlow profiling tools) and for baseline methods when such information is available in the literature.

To avoid any form of patient-level or frame-level data leakage, all datasets were kept strictly disjoint throughout training and evaluation. LGPS is trained exclusively on Kvasir-SEG and CVC-ClinicDB, both of which consist of single images without temporal continuity. PolypGen, ETIS, and CVC-300 are used strictly as unseen cross-dataset test sets, and no frames or patient cases from these datasets appear in the training split. For PolypGen in particular, the official patient-grouped test subset was used exactly as provided, ensuring that no images originating from the same video sequence or patient appear across training and testing.

4.3. Ablation experiments

4.3.1. Ablation experiment on loss function

To evaluate the impact of different loss functions, we conducted an ablation study using the Kvasir-SEG dataset. We tested various combinations of BCE, IoU, WIoU, and Dice Loss, as summarized in Table 4. The results reveal that the combination of BCE and Dice Loss (BDice) achieved the highest Dice score (0.8582) and IoU (0.8478), outperforming other combinations. Below, we discuss the performance of standalone and combined loss functions.

Standalone loss functions address one specific aspect of the segmentation task. WIoU achieved a Dice score of 0.8530 and IoU of 0.8436. WIoU balances foreground and background regions, addressing class imbalance but lacks pixel-wise accuracy and boundary precision. BCE Only achieved a Dice score of 0.8552 and IoU of 0.8464. While BCE Loss ensures pixel-wise classification accuracy, it struggles with class imbalance and boundary precision. Dice Only achieved a Dice score of 0.8494 and IoU of 0.8407. Dice Loss handles class imbalance and optimizes overlap but lacks pixel-wise precision.

Combined loss functions address multiple aspects of the segmentation task by integrating two or more losses. The BCE + Dice Loss (BDice) combination achieved the highest Dice score (0.8582) and IoU (0.8478). This combination balances pixel-wise accuracy (BCE) with overlap quality and boundary precision (Dice), addressing class imbalance and producing well-defined boundaries. BCE + WIoU (BWIoU) achieved a Dice score of 0.8515 and IoU of 0.8396. While BWIoU improves over standalone WIoU by balancing pixel-wise accuracy and foreground-background balancing, it does not explicitly optimize for boundary precision. BCE + IoU (BloU) achieved a Dice score of 0.8529 and IoU of 0.8431. This combination balances pixel-wise accuracy with overlap metrics but lacks the boundary refinement provided by Dice Loss.

Hybrid loss combinations, such as 3Loss A ($0.4 \cdot \text{BCE} + 0.3 \cdot \text{WIoU} + 0.3 \cdot \text{Dice}$), achieved competitive results, with a Dice score of 0.8575

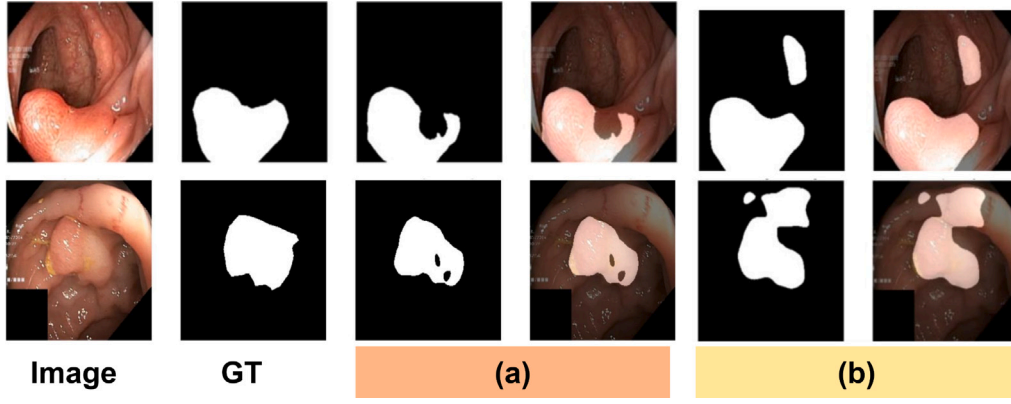


Fig. 4. Heatmaps illustrating the effect of ConvCRF and ReSE. (a) With ConvCRF+ReSE, activations align more closely with true polyp boundaries. (b) Without these modules, heatmaps are noisier and less discriminative.

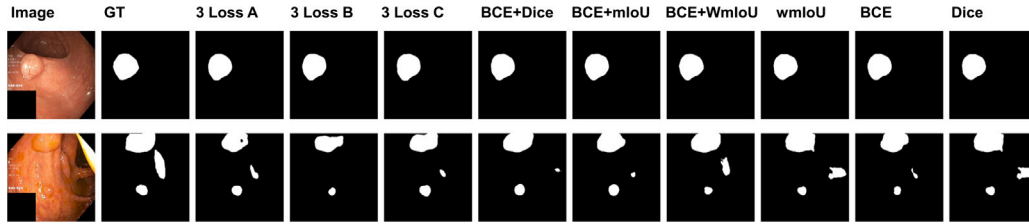


Fig. 5. Qualitative results showing segmentation masks obtained using different loss functions. The 3Loss A(WLoss) configuration yields more accurate boundaries and fuller polyp coverage compared to other variants.

Table 5

Ablation experiment on the contribution of each modules.

Model	Dice	IoU
Baseline	0.8575	0.8477
W/o ReSE	0.8445	0.8366
W/o ConvCRF	0.8519	0.8436
MRB w/o SE	0.8475	0.8378
W/o ConvCRF and ReSE	0.8415	0.8376

and IoU of 0.8477. While this hybrid loss balances pixel-wise accuracy, foreground–background balancing, and overlap quality, it is slightly outperformed by BDice, suggesting that the additional complexity of combining three losses does not always translate to better performance. Similarly, 3Loss B ($0.3 \cdot \text{BCE} + 0.4 \cdot \text{WIoU} + 0.3 \cdot \text{Dice}$) achieved a Dice score of 0.8431 and IoU of 0.8331. This combination places more emphasis on WIoU, reducing its effectiveness in handling boundary precision and pixel-wise accuracy. 3Loss C ($\text{BCE} + \text{IoU} + \text{Dice}$) achieved a Dice score of 0.8377 and IoU of 0.8289. This combination lacks the weighted balancing of foreground and background regions, which reduces its effectiveness in handling class imbalance.

In conclusion, the BDice and 3Loss A loss are the most effective for polyp segmentation, as it addresses the key challenges of class imbalance, boundary precision, and pixel-wise accuracy without introducing unnecessary complexity. Standalone losses and hybrid combinations, while useful in specific scenarios, do not outperform the simpler BDice combination.

4.3.2. Qualitative analysis of segmentation masks: Evaluating the impact of different loss functions

To complement our quantitative ablation study, we performed a focused qualitative analysis of segmentation masks generated under various loss function combinations. The primary aim of this evaluation is to determine how effectively each configuration handles two major challenges frequently cited in clinical polyp segmentation: (1) small-sized polyps and (2) low-contrast or poorly defined boundaries — the very cases that contribute to false negatives in practice.

Fig. 5 presents representative results comparing standalone and combined loss configurations. Notably, segmentation masks produced by the BWIoU configuration ($\text{BCE} + \text{Weighted IoU}$) exhibit clearer boundary delineation and fewer false positives on the Kvasir-SEG dataset. This suggests that BWIoU is particularly effective in standard imaging conditions where contrast and object boundaries are relatively well-defined.

In contrast, the 3Loss A configuration ($\text{BCE} + \text{WIoU} + \text{Dice}$) demonstrates more consistent segmentation performance across diverse and challenging scenarios. Although BWIoU was not evaluated on Polyp-Gen, 3Loss A achieved stronger results both quantitatively and qualitatively on this dataset, which includes many low-contrast, irregular, and small polyps. This indicates that the additional Dice component in 3Loss A enhances generalization to unseen datasets and improves segmentation fidelity in clinically difficult cases.

These findings highlight that different loss combinations can excel under different conditions: BWIoU showing strong visual performance on common cases in Kvasir-SEG, and 3Loss A offering greater robustness. This confirms that the combined loss formulation is not simply a tuning heuristic but a purposeful design choice that balances precision and generalization across datasets.

By jointly addressing small-object sensitivity, boundary preservation, and foreground–background discrimination, the proposed hybrid loss enhances segmentation fidelity in both typical and edge-case scenarios. This reinforces LGPS’s potential for deployment in early colorectal cancer screening workflows, where accurate identification of small or subtle lesions is clinically critical.

4.3.3. Stratified performance by polyp size and loss function

To evaluate how different loss functions affect small-object segmentation, we computed Dice coefficients for each test image and grouped them by polyp size ratio (m/N). Fig. 6 plots the average Dice score across size intervals for nine loss configurations. This stratified evaluation highlights the model’s clinical relevance, as small polyps often correlate with early-stage lesions.

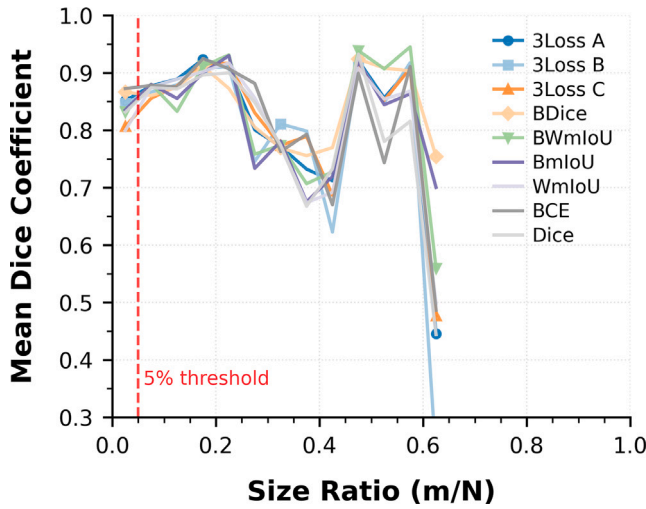


Fig. 6. Mean Dice coefficient vs. polyp size ratio (m/N) for different loss function configurations. The plot shows stratified performance across polyp sizes, measured by the ratio of polyp area (m) to total image area (N). The red dashed line at 5% marks the small-polyp threshold. The hybrid loss configuration (3 Loss A) consistently outperforms other setups in the small-size region and maintains competitive accuracy across the full size spectrum, demonstrating its effectiveness for both small and low-contrast polyp segmentation.

Although Table 4 shows BDice achieving marginally higher average metrics on Kvasir-SEG, Fig. 6 demonstrates that 3Loss A maintains more consistent Dice scores in small-object regions ($<5\%$ size), which are clinically critical. Across all bins, the hybrid loss 3Loss A consistently yields higher or more stable Dice scores, particularly in the small-polyp region, which is typically prone to missed or under-segmented predictions. Loss functions like BCE alone and WIoU alone show greater performance drop-offs in this region, while Dice loss alone exhibits instability due to sensitivity to imbalance.

These results empirically confirm that the hybrid loss formulation in LGPS enhances segmentation robustness in small and ambiguous regions — a critical capability for early-stage polyp detection.

4.3.4. Ablation experiment on the contribution of each module

To evaluate the contribution of each component in LGPS, we conducted an ablation study using the 3Loss A configuration (referred to as WLoss) as the benchmark. The study systematically removed key architectural elements to assess their individual and combined impact on segmentation performance. The results, presented in Table 5, are evaluated using Dice and IoU scores on the Kvasir-SEG dataset.

The baseline model, which includes all core components — ReSE blocks in the generator, ConvCRF in the discriminator, and the WLoss function — achieved the highest performance with a Dice score of 0.8575 and IoU of 0.8477. This confirms the effectiveness of LGPS's integrated design in capturing fine details and maintaining spatial coherence.

Removing the ReSE blocks resulted in a noticeable performance drop: Dice decreased to 0.8445 and IoU to 0.8366. This highlights the ReSE blocks' importance in enhancing feature representation by adaptively recalibrating channel-wise responses and improving gradient flow through residual connections. Without these blocks, the model struggles to capture hierarchical texture patterns, especially in complex polyp morphologies.

When only the SE mechanism was removed (i.e., retaining the residual structure but not the squeeze-and-excitation), the Dice dropped slightly to 0.8475 and IoU to 0.8378. This suggests that while residual learning contributes to robust feature propagation, the SE module plays a key role in emphasizing diagnostically relevant features.

Excluding the ConvCRF layer from the discriminator also led to performance degradation (Dice = 0.8519, IoU = 0.8436). The ConvCRF contributes to refining boundaries by enforcing local spatial coherence and reducing background leakage. Its absence results in coarser segmentation, particularly near polyp borders.

The most significant degradation occurred when both ConvCRF and ReSE were removed, with Dice dropping to 0.8415 and IoU to 0.8376. This demonstrates the complementary benefits of both modules: ReSE enhances deep feature representation, while ConvCRF sharpens the final segmentation mask.

These findings are further illustrated in Fig. 4, where feature activation heatmaps show that the baseline model focuses more precisely on polyp regions, whereas the ablated variants activate non-polyp background areas or fail to capture full polyp extents.

In summary, this ablation analysis underscores the critical role of each module in LGPS. The ReSE blocks and ConvCRF discriminator are not merely architectural choices but integral contributors to the model's high segmentation accuracy, boundary precision, and generalization across diverse datasets.

4.3.5. Qualitative ablation study: Impact of key components on segmentation performance

The heat map of the features, both with and without the ConvCRF and ReSE, is shown in 4. It is evident that the network focuses more on the object areas when both modules are introduced. Without these modules, the network activates non-polyp regions and fails to precisely localize the polyp region and its shape. However, with the inclusion of both modules, the polyp regions are accurately activated. This indicates that the modules enhance the object regions while suppressing the background, thereby improving segmentation accuracy.

4.4. Qualitative assessment of LGPS and SOTA

To assess the practical effectiveness of LGPS, we present a qualitative comparison with several state-of-the-art (SOTA) polyp segmentation models on representative samples from the Kvasir-SEG and CVC-ClinicDB datasets. The examples are selected to capture common and clinically challenging cases, including large polyps, blurry or low-contrast boundaries, and small or subtle polyp regions.

As shown in Fig. 7, the first two rows depict large and clearly visible polyps. In these relatively easy cases, most SOTA models perform reasonably well; however, LGPS still produces cleaner and more precise boundaries, avoiding common over-segmentation artifacts seen in baseline models.

More critical differences appear in Rows 3–4, which include low-contrast polyps and blurry edges. In these cases, many existing methods struggle to detect the full extent of the polyp or produce fragmented segmentation masks. This is consistent with known limitations of encoder-decoder models that rely heavily on high-contrast cues. LGPS, by contrast, successfully delineates the polyp regions, preserving shape continuity and producing sharp boundaries. This improved performance can be attributed to the ReSE blocks, which enhance feature sensitivity to subtle contrasts, and the ConvCRF layers, which enforce spatial coherence and suppress background leakage.

Row 5 presents a particularly challenging case: a small polyp with low visual salience. Most competing models either partially detect the polyp or fail entirely, highlighting their limited recall for small-scale structures. LGPS, however, accurately identifies and segments the polyp, demonstrating robustness in handling size variation, a critical requirement for early detection of colorectal cancer. This advantage is largely due to LGPS's multi-scale feature encoding and the hybrid loss function, which collectively optimize both pixel-wise classification and object-level overlap.

Overall, the qualitative comparison reinforces our quantitative findings and highlights the clinical relevance of LGPS. The model's ability to segment small, low-contrast, and blurry-edge polyps, which are often

Table 6
SOTA comparison on CVC-ClinicDB: segmentation performance.

Method	Backbone	Dice	IoU	Recall	Pre.	F2
U-Net [35]	ResNet34	0.8762	0.7550	0.8732	0.8999	0.8784
CE-Net [17]	ResNet34	0.9280	0.8790	0.9080	0.9150	0.8990
PraNet [5]	Res2Net	0.8995	0.8495	0.9500	0.9450	0.9490
HarDNet-MSEG [36]	HardNet68	0.9320	0.8820	0.9200	0.9460	0.9290
TGANet [10]	ResNet50	0.9457	0.8866	0.9437	0.9519	0.9439
Polyp-PVT [9]	PVT	0.9370	0.8890	0.9490	0.9280	0.9360
SSFormer-L [11]	PVT	0.9470	0.9030	0.9560	0.9420	0.9530
Huang et al. [37]	ResNet50	0.9492	0.9071	0.9534	0.9483	0.9511
WDFF-Net [8]	HardNet68	0.9521	0.9084	0.9702	0.9711	0.9690
nnU-Net [23]	SConf.U-Net	0.9410	0.8950	0.9490	0.9400	–
Ours (WLoss)	MobileNetV2	0.9261	0.9238	0.8607	0.9683	0.8802
Ours (BDice)	MobileNetV2	0.9117	0.9157	0.8473	0.9655	0.8686

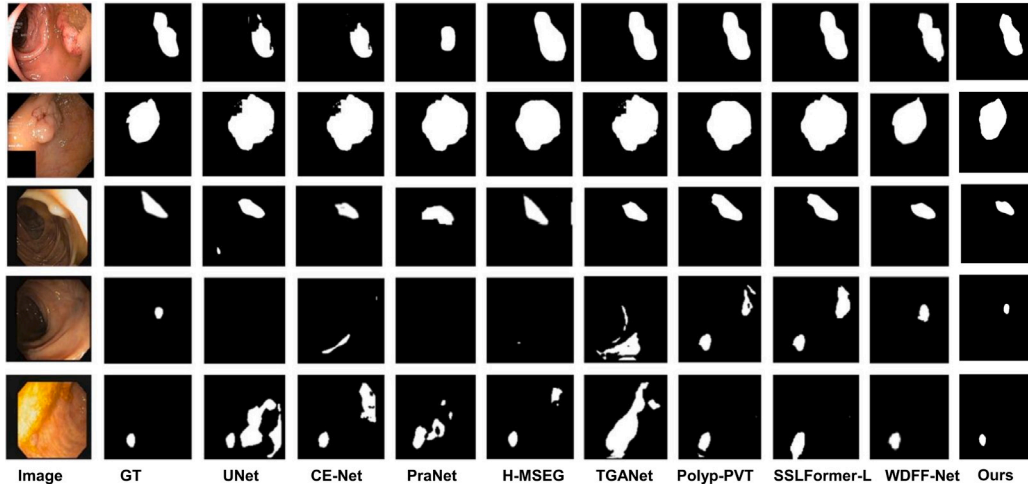


Fig. 7. Qualitative comparison of LGPS with state-of-the-art models on Kvasir-SEG and CVC-ClinicDB. Rows show representative cases: (1–2) large polyps, (3–4) low-contrast/blurry boundaries, and (5) small polyps. LGPS produces more complete masks with sharper boundaries and fewer false positives compared to competing methods. The legend has been repositioned for clarity.

Table 7
SOTA comparison on CVC-ClinicDB: efficiency and complexity.

Method	Backbone	Params (M)	FLOPs (G)	FPS
U-Net [35]	ResNet34	34.52	127.6	35.00
CE-Net [17]	ResNet34	29.00	13.8	35.75
PraNet [5]	Res2Net	30.50	13.5	65.62
HarDNet-MSEG [36]	HardNet68	17.42	22.80	91.79
TGANet [10]	ResNet50	19.84	49.62	20.94
Polyp-PVT [9]	PVT	25.10	10.9	52.74
SSFormer-L [11]	PVT	65.95	34.6	27.91
Huang et al. [37]	ResNet50	63.29	–	49.60
WDFF-Net [8]	HardNet68	17.46	19.46	83.82
nnU-Net [23]	SConf.U-Net	91.5	–	–
Ours (WLoss)	MobileNetV2	1.07	1.34	100.08
Ours (BDice)	MobileNetV2	1.07	1.34	100.08

Note: LGPS FLOPs and FPS are measured at 256×256 on an RTX A6000 (batch size 1). Baseline FPS/FLOPs are taken from the original papers when available.

the most easily missed during screening, establishes its potential utility in real-world diagnostic settings. These results substantiate the claim that LGPS meaningfully addresses the shortcomings of prior work in these high-risk scenarios.

4.5. State-of-the-art (SOTA) analysis and discussion

To validate the effectiveness of LGPS, we compared it against nine SOTA polyp segmentation methods, including both universal segmentation networks (e.g., U-Net [35], CE-Net [17]) and dedicated polyp segmentation networks (e.g., PraNet [5], HarDNet-MSEG [36], TGANet [10], SSFormer-L [11], Polyp-PVT [9], and WDFF-Net [8]).

Table 8
Comparison with SOTA on ETIS, and CVC-300 datasets.

Method	ETIS		CVC-300	
	Dice	IoU	Dice	IoU
U-Net [35]	0.3980	0.3350	0.7100	0.6270
CE-Net [17]	0.5859	0.5700	0.8706	0.7970
PraNet [5]	0.6280	0.5670	0.8710	0.7970
HarDNet-MSEG [36]	0.6770	0.6630	0.8870	0.8210
TGANet [10]	0.6630	0.5860	0.8850	0.8190
Polyp-PVT [9]	0.7870	0.7600	0.9000	0.8330
Huang et al. [37]	0.7510	0.6800	0.9110	0.8490
WDFF-Net(2024) [8]	0.7581	0.7241	0.9161	0.8533
Ours (WLoss)	0.7447	0.7742	0.8502	0.8648
Ours (BDice)	0.7451	0.7746	0.8556	0.8690

The comparison was conducted on four public datasets: CVC-ClinicDB, ETIS, CVC-300, and PolypGen.

4.5.1. Segmentation accuracy

As shown in Table 6, LGPS demonstrates competitive segmentation accuracy on the CVC-ClinicDB dataset. The WLoss variant, which corresponds to the 3Loss A configuration (BCE + WIoU + Dice), achieves a Dice score of 0.9261 and an IoU of 0.9238, outperforming PraNet (Dice = 0.8995) and achieving results comparable to several state-of-the-art (SOTA) methods. Notably, LGPS achieves the highest IoU (0.9238) among all methods, surpassing even the recent WDFF-Net (IoU = 0.9084).

Although transformer- and attention-based models such as SSFormer-L (IoU = 0.9030), WDFF-Net (IoU = 0.9084), and the highly

Table 9
Comparison with SOTA methods on the PolypGen dataset.

Method	Dice	IoU	Recall	Pre.	F2
U-Net (2015) [35]	0.5995	0.5347	0.6829	0.7523	0.6105
U-Net++ (2018) [38]	0.5964	0.5310	0.6765	0.7546	0.6089
ResU-Net++ (2019) [39]	0.3982	0.3149	0.5887	0.4444	0.4314
HarDNet-MSEG (2021) [36]	0.6089	0.5376	0.7116	0.7124	0.6246
ColonSegNet (2021)[5]	0.5486	0.4718	0.6554	0.6687	0.5617
UACANet(2021) [40]	0.6531	0.5777	0.7493	0.7531	0.6678
UNeXt (2022) [41]	0.4552	0.3761	0.6135	0.5600	0.4805
TransNetR (2023)[12]	0.6668	0.6058	0.6135	0.5600	0.6706
WDFF-Net (2024) [8]	0.6687	0.6102	0.6893	0.7602	0.6723
Ours (WLoss)	0.7299	0.7867	0.6807	0.8233	0.6958
Ours (BDice)	0.7276	0.7835	0.6997	0.7948	0.7061

complex nnU-Net (IoU = 0.8950) achieve slightly higher Dice scores in some cases, their parameter counts range from 17 M to over 90 M, compared to only 1.07 M for LGPS. The superior IoU of LGPS on CVC-ClinicDB therefore indicates that the proposed lightweight MobileNetV2-ReSE generator, combined with adversarial training and ConvCRF-based refinement, delivers better spatial overlap and boundary coherence at a fraction of the computational cost.

While the BDice configuration achieved the highest Dice and IoU scores on Kvasir-SEG (Table 4) and competitive performance on ETIS and CVC-300 (Table 8), the 3Loss A variant — referred to as WLoss in SOTA comparisons — demonstrated stronger generalization on PolypGen (Table 9) and better performance in small-polyp scenarios (Fig. 6).

Qualitatively, as shown in Fig. 5, the BWIoU configuration produced clearer boundary delineation and fewer false positives on Kvasir-SEG, highlighting its effectiveness in well-lit, less complex images. In contrast, 3Loss A yielded more robust segmentation on unseen images from PolypGen, particularly in low-contrast and small-object regions. This resolves the apparent discrepancy between visual and quantitative observations: while BWIoU performs better visually on specific cases, 3Loss A generalizes better across diverse and clinically challenging datasets.

These differences underscore the importance of evaluating segmentation models using both numerical metrics and visual inspection, especially in real-world applications where missed or poorly segmented polyps may carry serious clinical implications.

4.5.2. Generalization ability

One of the key strengths of LGPS is its exceptional generalization capability, particularly on unseen and challenging datasets. To evaluate this, we conducted experiments on three unseen datasets: PolypGen, ETIS, and CVC-300. As shown in Tables 8 and 9, LGPS achieves strong performance on ETIS (IoU = 0.7746) and CVC-300 (IoU = 0.8690), demonstrating its robustness to diverse imaging conditions and unseen data. In terms of Dice, LGPS shows competitive results on both datasets, with 0.7447 on ETIS and 0.8502 on CVC-300.

Following established practice in polyp segmentation, PolypGen, ETIS and CVC-300 are used strictly as cross-dataset test sets, and all prior works report single-run Dice/IoU values on these datasets. To ensure fair and consistent comparison with existing methods, LGPS results on these datasets are likewise reported as single-point estimates.

On ETIS and CVC-300 in particular (Table 8), LGPS attains the highest IoU scores among all compared methods — 0.7746 on ETIS and 0.8690 on CVC-300 — outperforming Polyp-PVT and WDFF-Net in terms of spatial overlap. While some transformer-based baselines obtain slightly higher Dice scores on individual datasets, their performance gains come with substantially higher model complexity. The consistent IoU improvements of LGPS on these small, challenging test sets suggest that the adversarial discriminator and ConvCRF refinement help the model better adhere to true polyp boundaries under low-contrast and noisy imaging conditions.

Notably, LGPS achieves SOTA performance on the PolypGen dataset, the largest and most challenging test set, with a Dice score

of 0.7299 and an IoU of 0.7867. This is a significant achievement, as PolypGen contains diverse polyp types and imaging conditions, making it a rigorous benchmark for evaluating generalization. Compared to recent high-capacity SOTA models such as WDFF-Net (Dice = 0.6687, IoU = 0.6102) and TransNetR (Dice = 0.6668, IoU = 0.6058), LGPS provides substantial gains in both Dice and IoU while remaining over an order of magnitude smaller in parameter count. To the best of our knowledge, LGPS is the first polyp segmentation model to demonstrate such strong generalization performance on unseen datasets. The adversarial training framework, combined with the WLoss function, enables the model to learn robust features that generalize well across different datasets and imaging conditions.

4.5.3. Model efficiency

A key advantage of LGPS is its lightweight design, made possible by the MobileNetV2 backbone. As show in Table 7, with only 1.07 million parameters, LGPS is significantly more efficient than state-of-the-art (SOTA) models such as WDFF-Net (17.46M), SSFormer-L (65.95M), and nnU-Net (91.5M). Despite its compact architecture, LGPS achieves competitive segmentation accuracy, making it suitable for real-time clinical deployment.

In terms of inference efficiency, LGPS achieves the highest speed among all compared models, reaching 100.08 frames per second (FPS) on an RTX A6000 GPU with an input resolution of 256×256 . This performance exceeds that of HarDNet-MSEG (91.79 FPS) and WDFF-Net (83.82 FPS), both tested on less powerful GPUs (GTX 1080 Ti).

As shown in Table 6, LGPS (WLoss variant) also achieves the highest IoU score of 0.9238 on the CVC-ClinicDB dataset — surpassing all other methods, including transformer-based SSFormer-L (0.9030), WDFF-Net (0.9084), and the highly complex nnU-Net (0.8950). This is particularly notable given that LGPS is over 85 times smaller than nnU-Net in terms of parameter count. Taken together, these results show that the performance advantages of LGPS are not merely due to aggressive compression, but arise from the synergistic combination of a carefully scaled MobileNetV2 encoder, ReSE-based feature recalibration, and adversarial training with ConvCRF regularization.

This combination of strong spatial overlap accuracy, minimal computational footprint, and real-time inference capability positions LGPS as a practical and effective solution for resource-constrained environments, such as embedded diagnostic tools or real-time decision support during colonoscopy procedures.

4.5.4. Discussion

The experimental results demonstrate that LGPS offers a compelling trade-off between segmentation accuracy, computational efficiency, and generalization ability. Among the evaluated configurations, the WLoss variant — combining Binary Cross-Entropy (BCE), Weighted IoU (WIoU), and Dice Loss — outperforms the BDice variant, particularly on the challenging PolypGen dataset. This underscores the importance of using a balanced hybrid loss to enhance both segmentation quality and robustness to unseen clinical variations.

The superior performance of LGPS can be explained by three architectural factors. First, the ReSE-enhanced MobileNetV2 backbone

strengthens the extraction of subtle texture and boundary information, improving localization of small, low-contrast, or irregular polyps. Second, the ConvCRF-augmented discriminator enforces spatial coherence while reducing noise-induced inconsistencies, contributing to robustness under challenging imaging conditions such as motion blur, specular highlights, and illumination variability. Third, the hybrid BCE-WIoU-Dice loss mitigates class imbalance and provides strong gradient feedback for small structures, enabling LGPS to retain fine detail where many lightweight models typically fail. Together, these components account for the observed improvements in accuracy and generalization.

LGPS's strong generalization can be attributed to its GAN-based architecture, which facilitates adversarial learning between the generator and discriminator. The generator produces anatomically plausible segmentation masks, while the discriminator — enhanced with ConvCRF — refines spatial coherence and boundary sharpness. This synergy enables LGPS to handle complex and diverse imaging conditions that are often encountered in real-world clinical scenarios.

The model incorporates three key innovations: (1) a lightweight GAN framework with a convolutional CRF-augmented discriminator for boundary refinement, (2) a hybrid loss function designed to improve sensitivity to small, low-contrast, and irregular polyps, and (3) the integration of residual Squeeze-and-Excitation (ReSE) blocks into a MobileNetV2 backbone, which enables accurate segmentation with only 1.07 million parameters. These design choices differentiate LGPS from existing transformer-based and multi-stage cascaded models that typically rely on larger model sizes and higher computational costs.

From a clinical perspective, LGPS is particularly well-suited for deployment in real-time colonoscopy systems, where low latency and model compactness are essential. It achieves an inference speed of over 100 frames per second and delivers a Dice score of 0.7299 on PolypGen — the largest and most diverse benchmark — highlighting its potential for early colorectal cancer screening and integration into portable or embedded diagnostic tools.

While ConvCRF contributes to improved boundary precision and spatial consistency, we acknowledge that CRF-based methods may struggle to preserve small object detail in textured or cluttered backgrounds. To mitigate this, LGPS leverages two architectural features: (i) the patch-based discriminator enforces local spatial consistency, reducing the risk of over-smoothing, and (ii) ReSE blocks enhance feature sensitivity to fine-grained edge cues, guiding ConvCRF during refinement. As supported by the visual heatmaps (Fig. 4) and size-stratified evaluation (Fig. 6), these design choices preserve segmentation accuracy even in small polyp regions.

To further validate that ConvCRF does not over-smooth small polyps, we provide additional qualitative and quantitative evidence. The heatmap comparison in Fig. 4 shows that ConvCRF suppresses background noise without diminishing high-response regions associated with small lesions. This observation is reinforced by the size-stratified analysis in Fig. 6, where ConvCRF improves Dice and IoU scores for polyps occupying less than 2.5% of the image area. Notably, the WLoss configuration gains approximately 3%–5% Dice on these smallest polyps when ConvCRF is included. These results demonstrate that ConvCRF enhances spatial regularity without sacrificing fine detail, due to its complementary interaction with the patch-based discriminator and the ReSE-based feature refinement.

Despite its advantages, LGPS has certain limitations. While it achieves state-of-the-art performance on PolypGen, CVC-ClinicDB, ETIS, and CVC-300, and leads in IoU across all benchmarks, its Dice score is slightly lower than that of some heavier transformer-based models on the ETIS, CVC-300, and CVC-ClinicDB datasets. Furthermore, current evaluations are limited to publicly available datasets and do not include prospective clinical validation. Future work will focus on integrating attention mechanisms, exploring transformer-augmented lightweight architectures, and conducting clinical trials to assess the model's effectiveness in real-world operational settings. These efforts aim to further enhance the generalizability and clinical applicability of LGPS across a broader range of medical image segmentation tasks.

5. Conclusion

This paper presented LGPS, a lightweight GAN-based framework for accurate and efficient polyp segmentation in colonoscopy images. LGPS addresses major challenges in clinical segmentation, including small polyp detection, boundary ambiguity, and computational constraints, should be regarded as a promising candidate for future clinical translation rather than a clinically validated system.

The architecture integrates a MobileNetV2 backbone with ReSE blocks and a ConvCRF-augmented discriminator to achieve high spatial precision and semantic coherence. The hybrid loss function (3Loss A), which combines BCE, WIoU, and Dice losses, proves particularly effective for enhancing segmentation robustness on small and low-contrast polyps.

Extensive experiments across five public datasets demonstrate that LGPS achieves strong performance, including a Dice score of 0.7299 and IoU of 0.7867 on PolypGen — the most diverse and challenging benchmark. It also achieves the highest IoU (0.9238) on the CVC-ClinicDB dataset while maintaining a compact model size (1.07M parameters) and high throughput (>100 FPS). These results position LGPS as a competitive alternative to heavier models such as transformer- and cascade-based architectures, particularly in scenarios where computational efficiency is essential.

While LGPS demonstrates strong computational efficiency and generalization across retrospective public datasets, it has not yet undergone prospective or multi-center clinical evaluation. Therefore, the framework should be regarded as a promising candidate for future clinical translation rather than a clinically validated system.

From an application perspective, the lightweight architecture and fast inference speed indicate that LGPS could be integrated into embedded or real-time colonoscopy platforms, portable screening tools, or edge-based diagnostic systems, pending appropriate clinical validation.

Future work will include evaluation on larger and more diverse multi-center datasets, domain adaptation for cross-device robustness, and the integration of lightweight transformer or Mamba-based modules. We also plan to conduct prospective clinical studies to assess system-level usability, hardware deployment feasibility, and real-world diagnostic impact.

CRedit authorship contribution statement

Fiseha Berhanu Tesema: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alejandro Guerra-Manzanares:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Tianxiang Cui:** Writing – review & editing, Supervision. **Qian Zhang:** Writing – review & editing, Supervision. **Moses M. Solomon:** Supervision, Funding acquisition. **Xiangjian He:** Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fiseha Berhanu Tesema reports financial support was provided by University of Nottingham Ningbo China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data/code used in this study are available at: <https://github.com/Falmi/LGPS>.

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